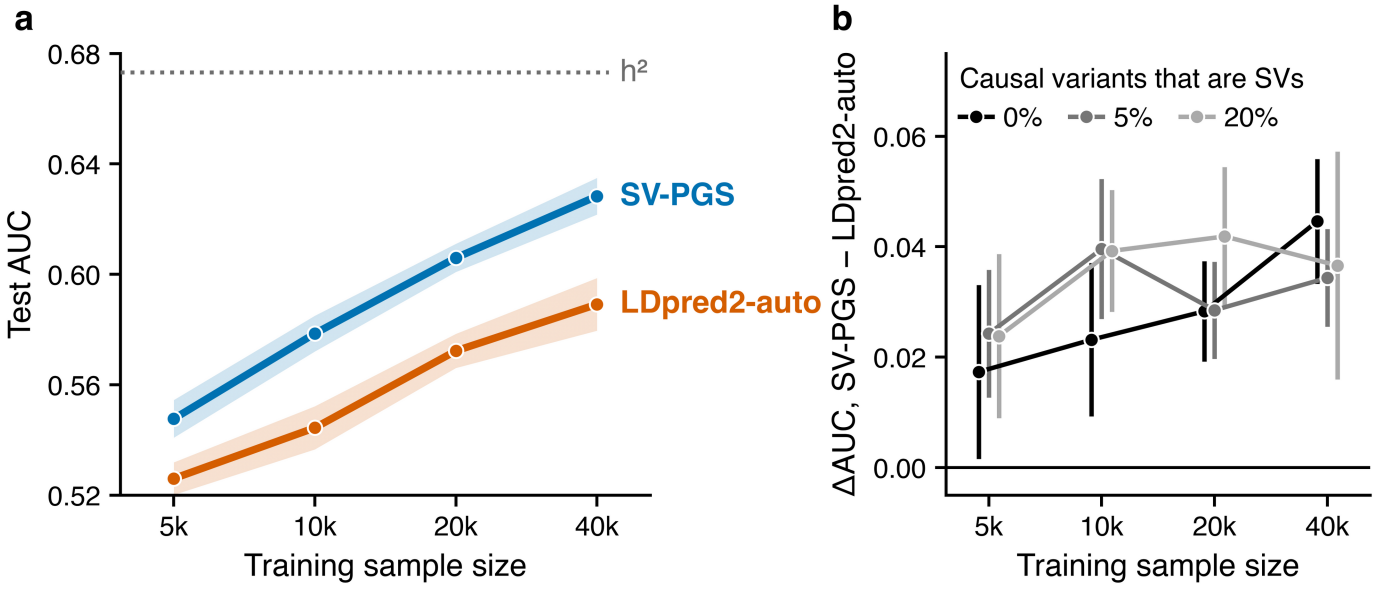


Figure 1 | SV-PGS gains accuracy faster than LDpred2-auto as the training sample grows, by the same margin whether 0%, 5% or 20% of the causal variants are SVs. A simulated binary trait on chromosome 22 (liability $h^2 = 0.1$, prevalence 10%, 200 causal variants, SV effects drawn like small-variant effects), scored in the same 10,000 held-out people by SV-PGS fitted on every SNV, indel and imputed SV and by LDpred2-auto fitted on HapMap3-density SNVs of the same training people. a, Test AUC, mean and Student-t 95% interval over the replicates of all three settings pooled (42, 37, 38, 16 replicates at 5k, 10k, 20k and 40k). Dotted: the AUC of a score that captures all of h^2 (Wray et al. 2010), the most any method can reach. b, The paired difference in AUC between the methods on the same held-out people for each setting separately, mean and Student-t 95% interval over its replicates; the three overlap at every size, which is why a pools them. bench-sim v7 cohort from 1000 Genomes haplotypes, Beagle-imputed SVs; fitted on MSI.

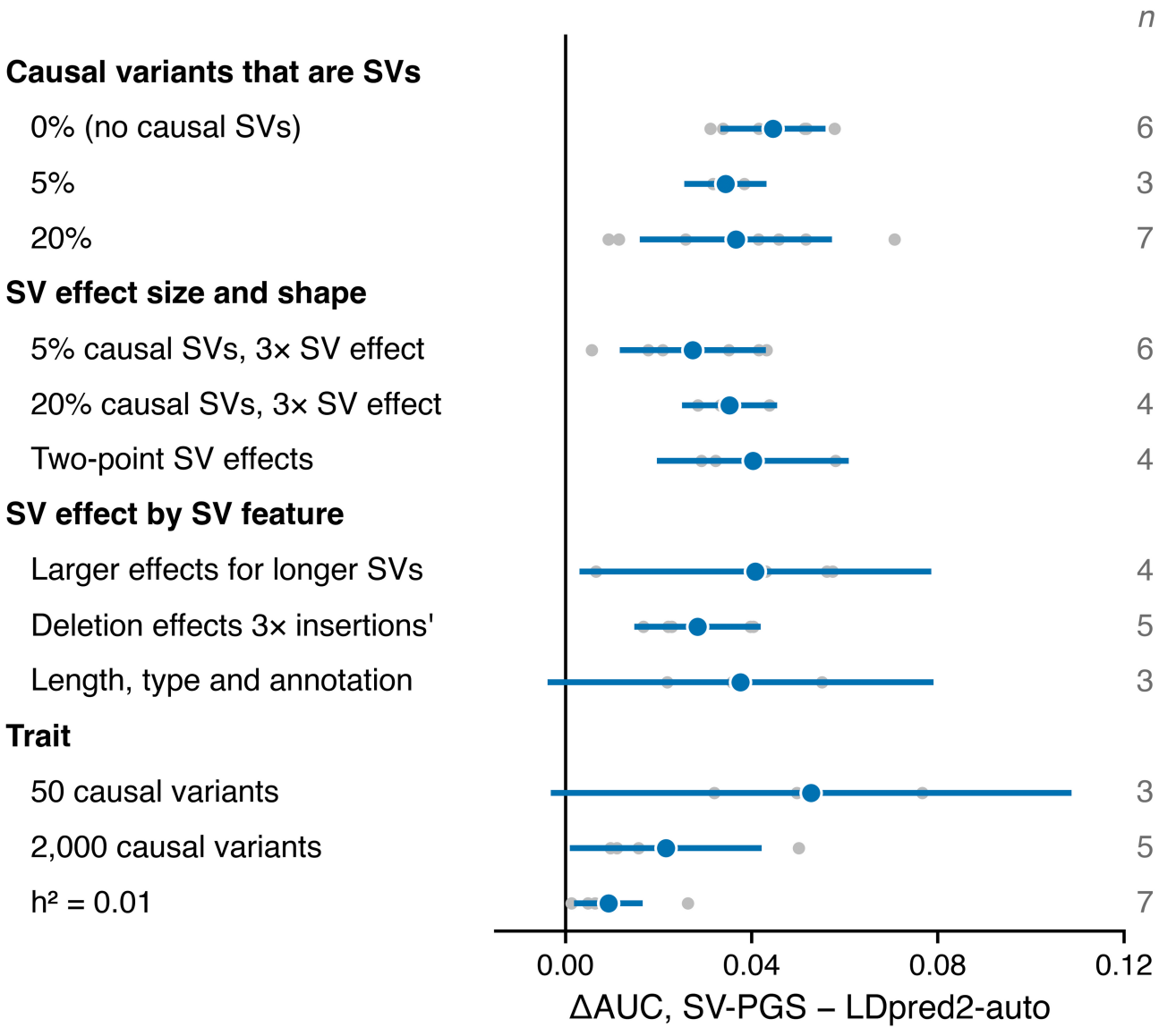


Figure 2 | SV-PGS is more accurate than LDpred2-auto across simulated genetic architectures. Paired difference in test AUC on the same 10,000 held-out people between SV-PGS (every SNV, indel and imputed SV) and LDpred2-auto (HapMap3-density SNVs), both trained on the same 40,000 people: replicates (grey), mean and Student-t 95% interval (blue); n replicates at right. SV-PGS had the higher AUC in 57 of 57 replicates. Unless named, an architecture has liability $h^2 = 0.1$, prevalence 10%, 200 causal variants and 20% causal SVs, with SV effects drawn from the same distribution as small-variant effects; '3× SV effect' triples the SV effect SD, 'well-imputed' draws causal SVs only from those Beagle imputes with $r^2 \geq 0.8$ and 'common' only from SVs of frequency ≥ 5%. Not yet replicated three times at 40,000: heavy-tailed sv effects, only well-imputed svs, only well-imputed svs, 3× effect, only common svs, prevalence 1%.

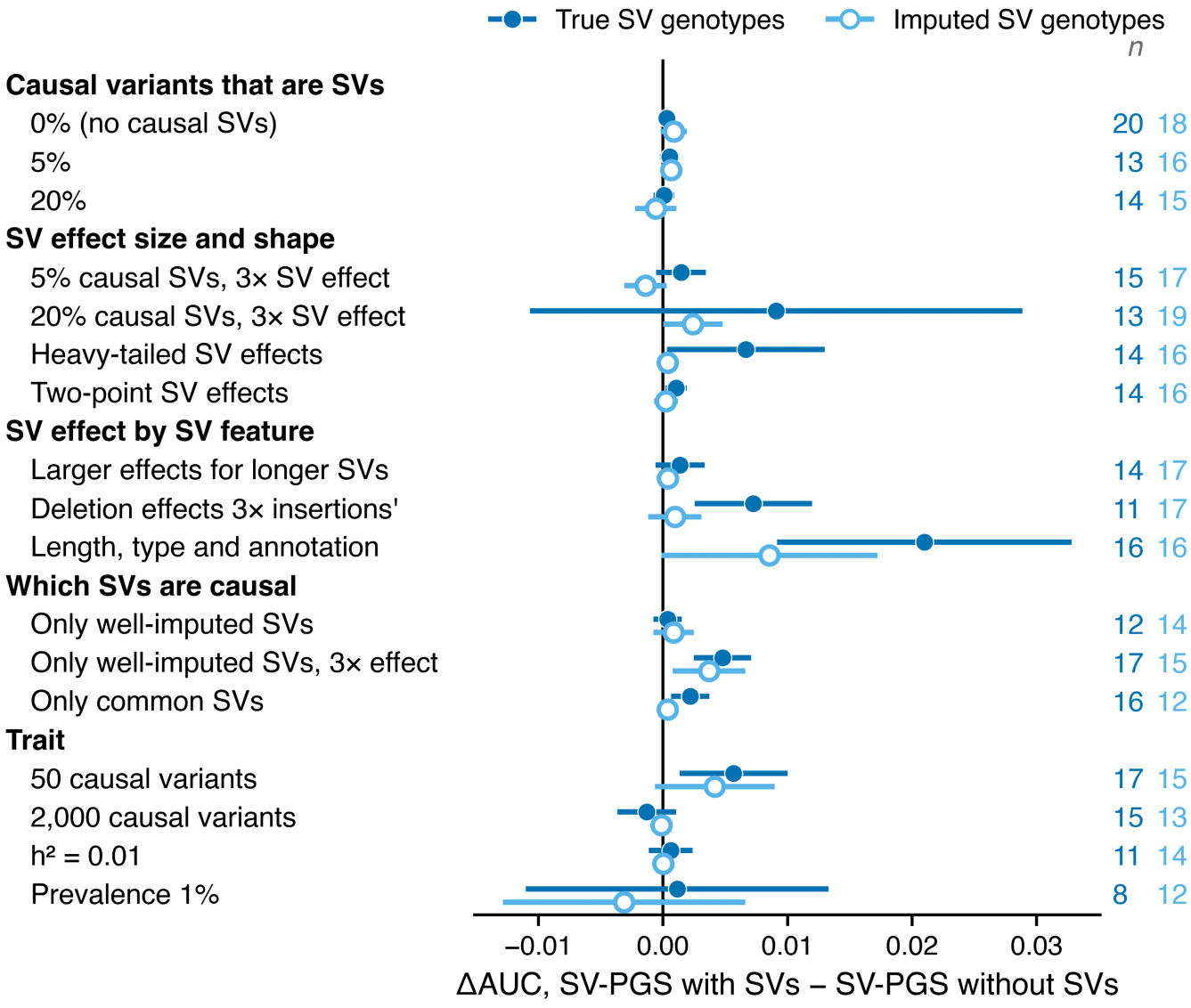


Figure 3 | SVs add accuracy when they are measured well; imputation removes most of the gain. Paired difference in test AUC on the same held-out people between SV-PGS fitted with and without SVs, when every SV is measured by its true genotype (dark) or imputed by Beagle from SNV haplotypes (light), on one 3 Mb chromosome 22 window (60,000 records). Means with Student-t 95% intervals over replicates, training sizes 5,000-40,000 pooled; n replicates at right (true, imputed). The true-genotype interval lies above zero for 7 of 17 architectures. Same simulated people, truth and fits; only the SV genotypes differ. Unless named, an architecture has liability $h^2 = 0.1$, prevalence 10%, 200 causal variants and 20% causal SVs, with SV effects drawn from the same distribution as small-variant effects; '3× SV effect' triples the SV effect SD, 'well-imputed' draws causal SVs only from those Beagle imputes with $r^2 \geq 0.8$ and 'common' only from SVs of frequency ≥ 5%.

a	True SV genotypes			b Imputed SV genotypes		
	n = 5k	n = 10k	n = 20k	n = 5k	n = 10k	n = 20k
Causal variants that are SVs						
0% (no causal SVs)	0.000	**<0.001**	0.000	+0.001	0.000	+0.001
5%	+0.001	+0.001	+0.001	0.000	+0.001	+0.001
20%	0.000	0.000	+0.001	0.000	−0.002	+0.002
SV effect size and shape						
5% causal SVs, 3× SV effect	0.000	0.000	+0.005	−0.002	−0.001	−0.001
20% causal SVs, 3× SV effect	**+0.013**	+0.007	–	+0.002	+0.003	+0.003
Heavy-tailed SV effects	+0.006	+0.008	+0.007	+0.001	0.000	0.000
Two-point SV effects	0.000	+0.001	+0.002	0.000	0.000	+0.001
SV effect by SV feature						
Larger effects for longer SVs	0.000	+0.001	+0.002	0.000	+0.002	0.000
Deletion effects 3× insertions'	+0.003	+0.007	**+0.010**	0.000	0.000	+0.003
Length, type and annotation	+0.026	+0.018	+0.022	+0.015	+0.005	+0.007
Which SVs are causal						
Only well-imputed SVs	+0.001	0.000	–	0.000	+0.002	−0.001
Only well-imputed SVs, 3× effect	+0.002	**+0.007**	+0.004	+0.002	+0.005	–
Only common SVs	0.000	**+0.003**	0.000	**<0.001**	0.000	+0.001
Trait						
50 causal variants	+0.009	+0.003	+0.005	+0.010	+0.003	+0.001
2,000 causal variants	−0.003	−0.002	0.000	0.000	−0.001	0.000
$h^2 = 0.01$	0.000	+0.002	0.000	0.000	+0.001	0.000
Prevalence 1%	+0.002	−0.004	+0.006	0.000	−0.007	−0.002

Figure 4 | The gain from adding SVs by architecture and training size. Mean paired difference in test AUC on the same held-out people between SV-PGS fitted with and without SVs on one chromosome 22 window, with SVs measured by their true genotypes (a) or imputed (b). Bold on full colour: the Student-t 95% interval over replicates excludes zero ('<0.001': a gain below 0.001 whose interval excludes zero); grey on pale colour: it does not. Colour saturates at ±0.013. Dashes: fewer than two replicates. 40,000-person window fits are not yet replicated and are left out. Unless named, an architecture has liability h^2 = 0.1, prevalence 10%, 200 causal variants and 20% causal SVs, with SV effects drawn from the same distribution as small-variant effects; '3× SV effect' triples the SV effect SD, 'well-imputed' draws causal SVs only from those Beagle imputes with $r^2 \geq 0.8$ and 'common' only from SVs of frequency ≥ 5%.

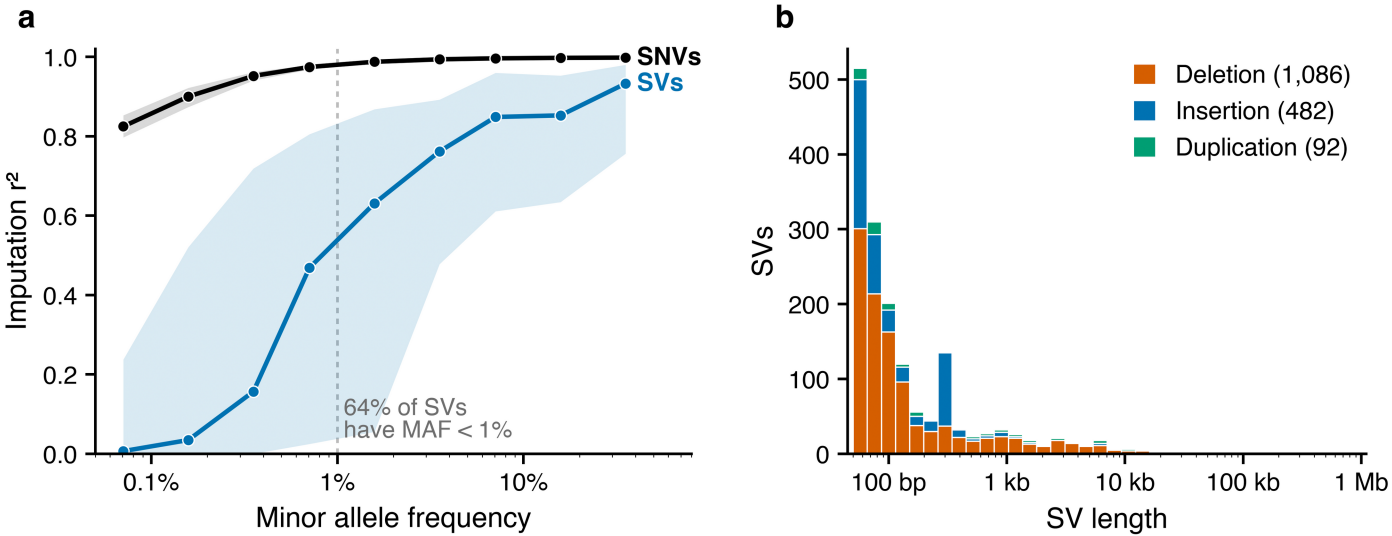


Figure 5 | The simulated SVs are mostly rare and short, and rare SVs impute poorly. a, Dosage r² between Beagle-imputed and true genotypes of 1,662 SVs and 6,000 sampled SNVs on chromosome 22 in 50,000 simulated people, by minor allele frequency: medians (points) and interquartile ranges (bands) per frequency bin. b, Lengths of the 1,662 measured SVs, stacked by class (sequence-resolved records without a type are classed by the sign of their length change). bench-sim v7 cohort from public 1000 Genomes haplotypes.

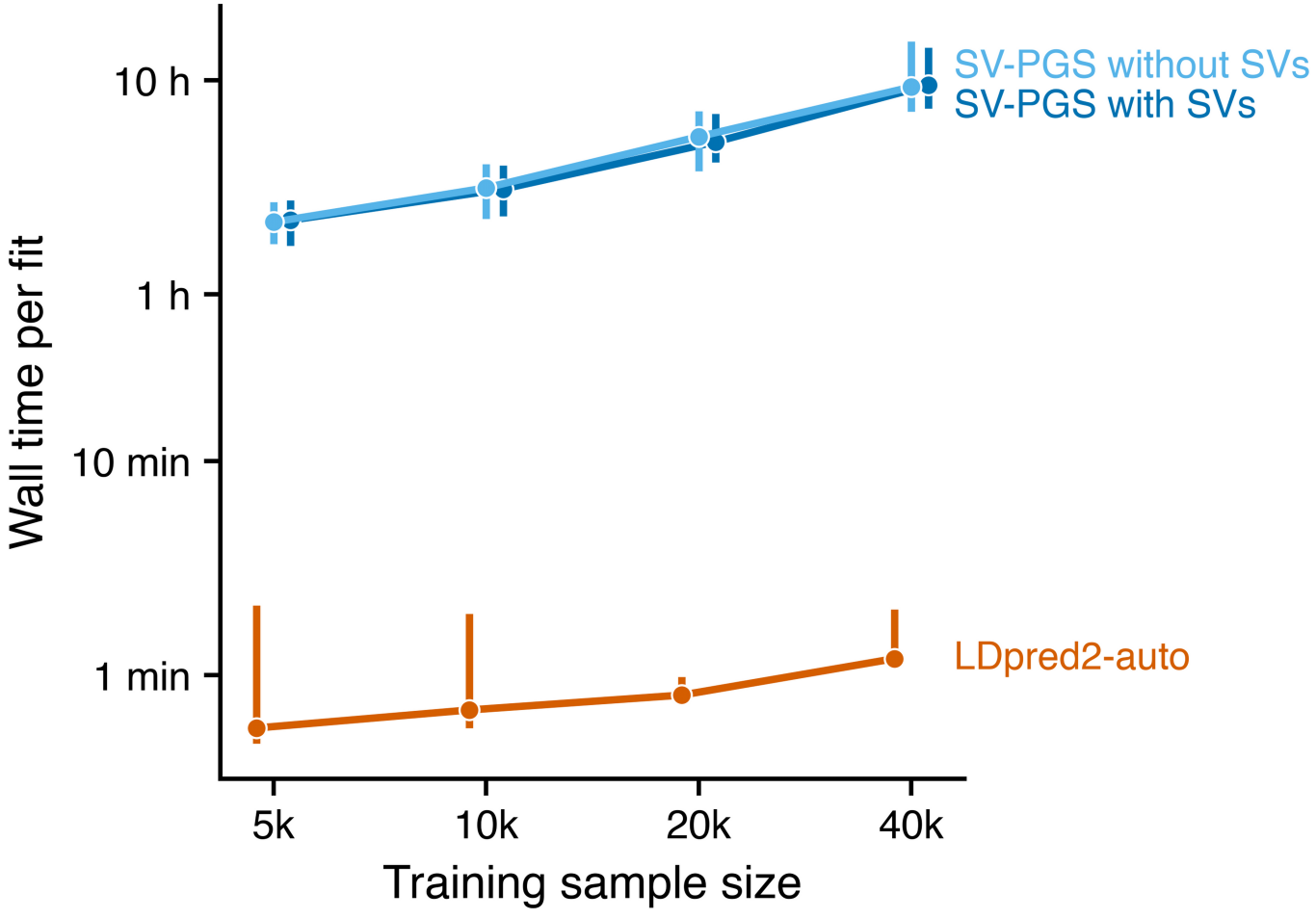


Figure 6 | Fit time. Wall time of one fit on chromosome 22 per method and training size, pooled over every architecture: median (points) and interquartile range (bars). MSI Slurm jobs, one fit per job; SV-PGS fits every variant by empirical Bayes, LDpred2-auto HapMap3-density SNVs.